

Benchmarkate

Your AI labmate

I'm a scientist



Every step of the scientific process.

ASK

HYPOTHESIZE

EXPERIMENT

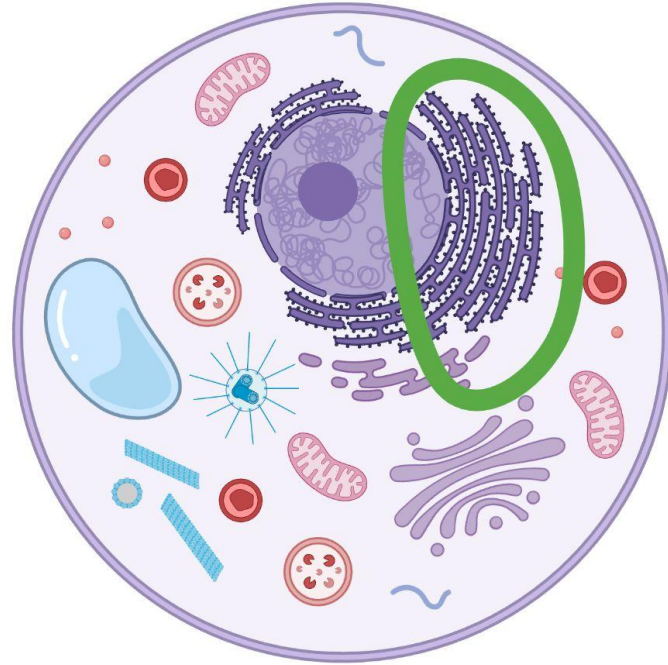
ITERATE

What is already known?

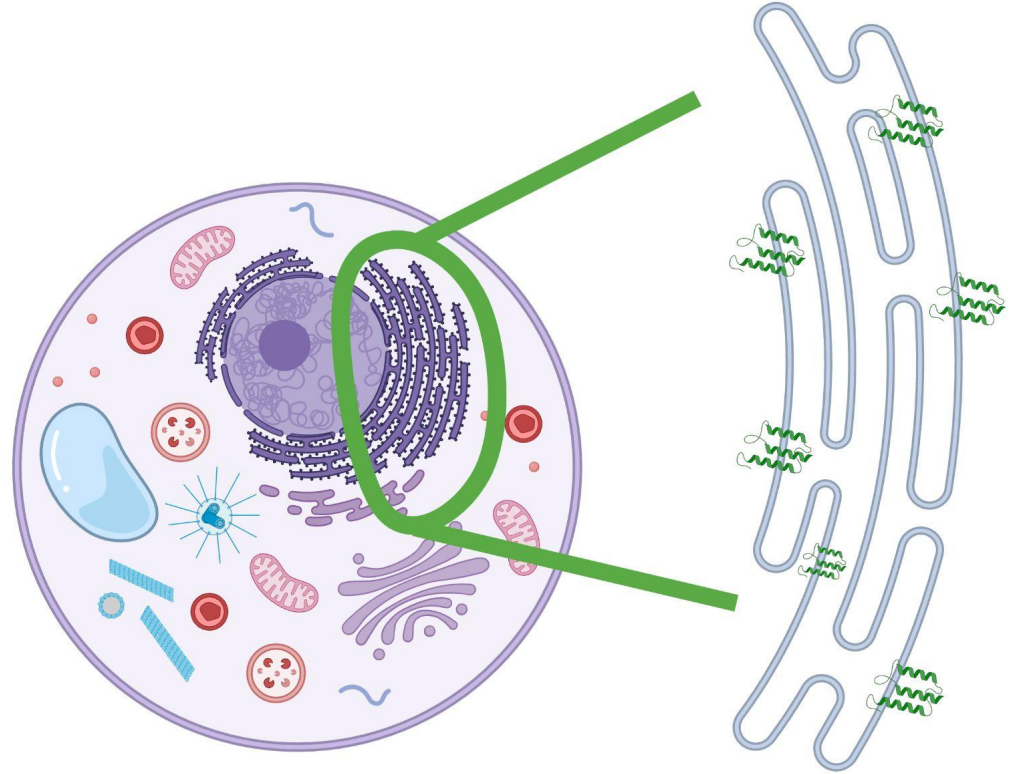
ASK

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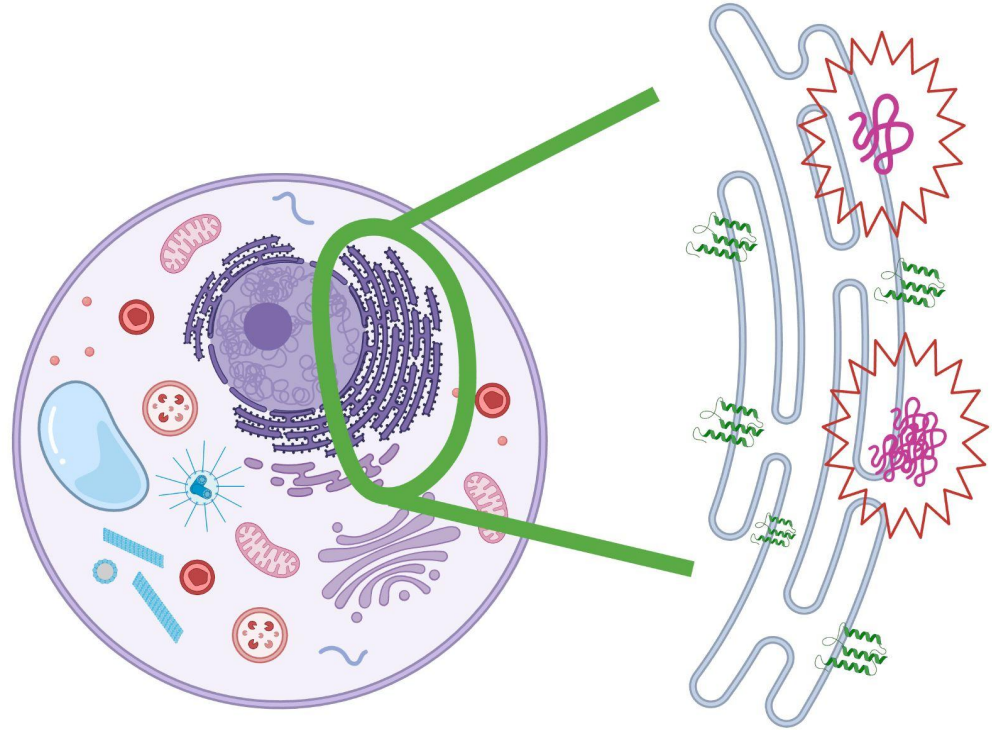
ERAD



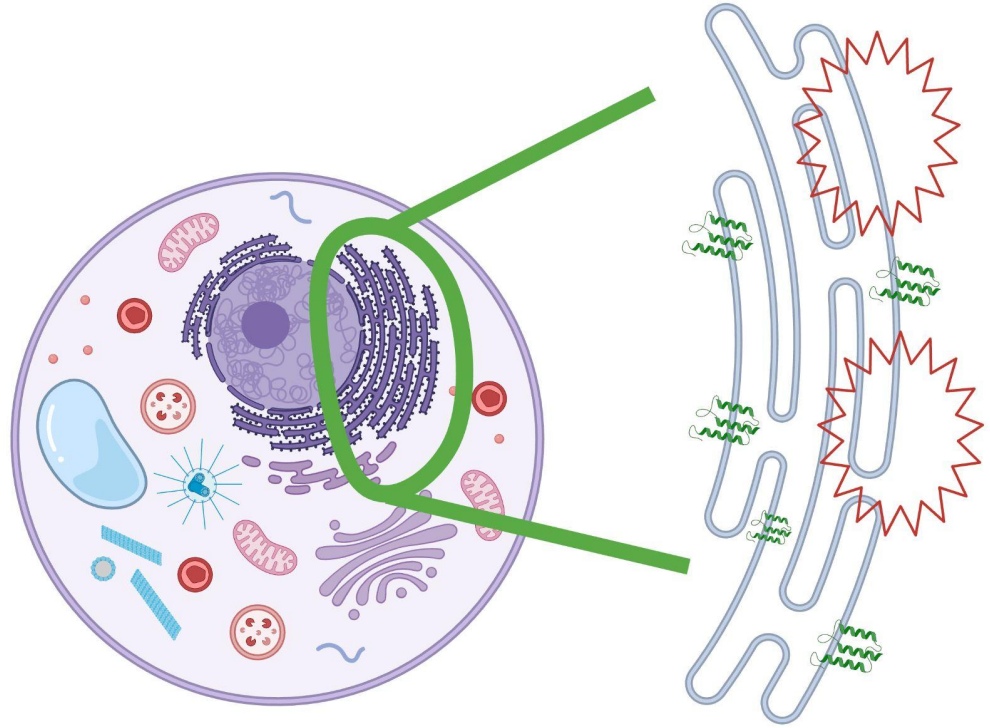
ERAD



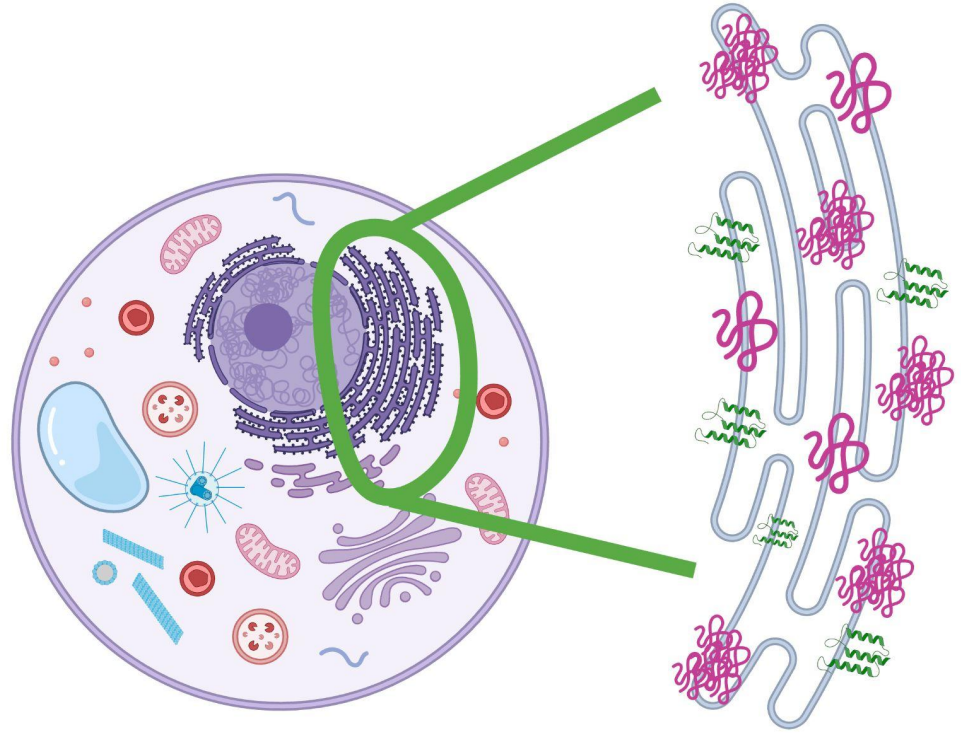
ERAD



ERAD



~~ERAD~~



“What ERAD genes drive bortezomib resistance in multiple myeloma?”

Tip: The Start here tab can fill these in for a question

Antibody-secreting plasma cells live at the edge of the UPM. If your gene's loss-of-function produces an ERAD or proteotoxic phenotype, the signal will be biggest here.

Generate Cellb notebook

Resolved 4 gene(s) for context: Plasma cells (extreme ER stress)

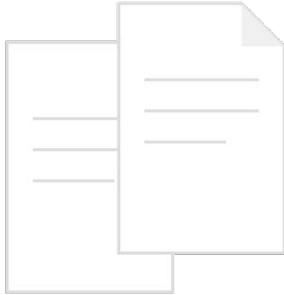
```
{
  "HSP45" : "ENSG00000044574"
  "SEL1L" : "ENSG00000067153"
  "DCRL1" : "ENSG00000136666"
  "UBE2J1" : "ENSG00000106855"
}
```

Notebook saved.

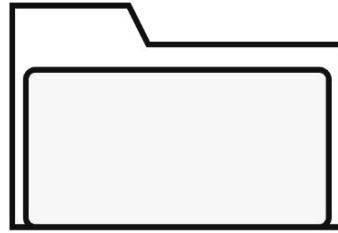
notebooks/generated/perturb_HSP45_SEL1L_UBE2J1_plasma_cells_extreme_er_stress_835feb12web

ASK

PUBMED



PMID 31368616
PMID 24069311



Evidence relevant to
my question

GENEFORMER



Seven agents, one goal.

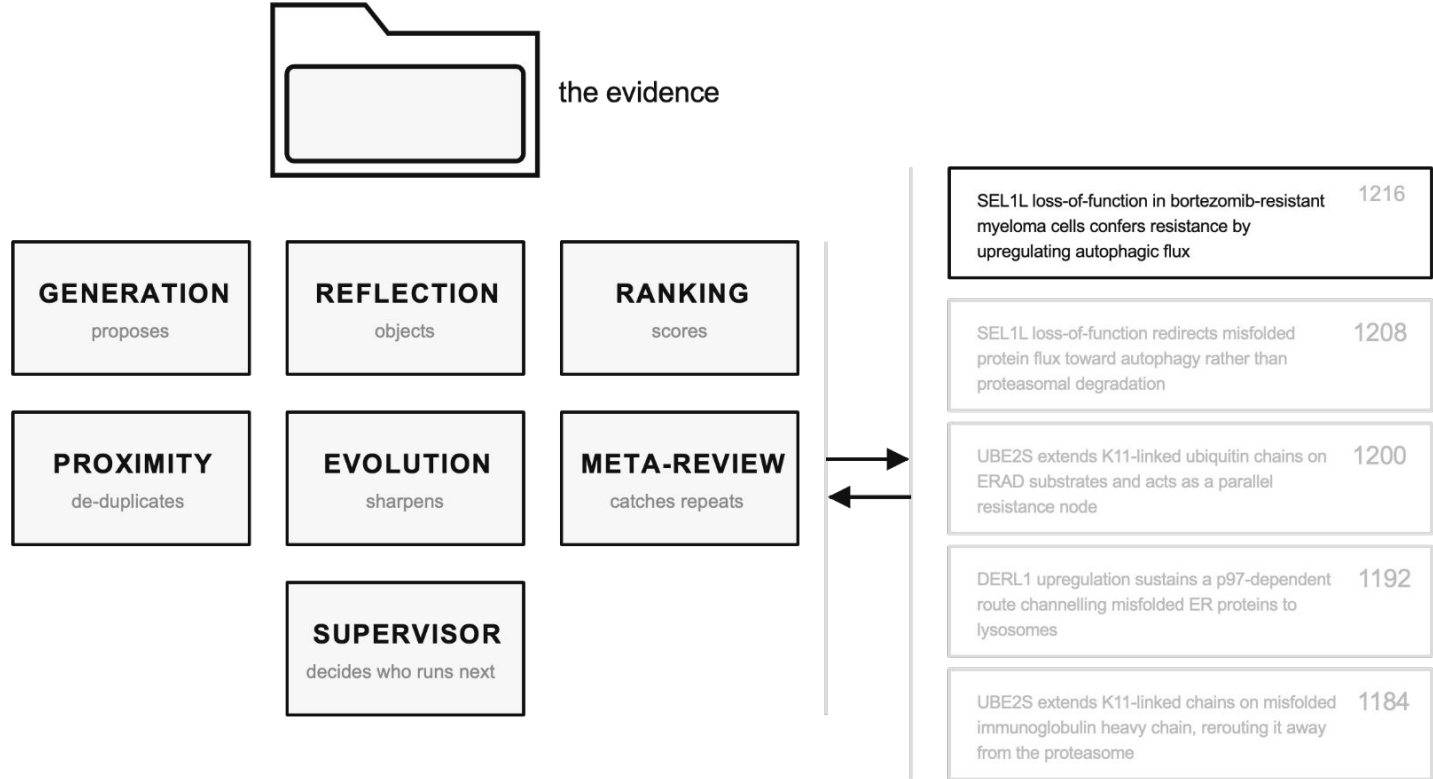
HYPOTHESIZE

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Seven agents to generate hypotheses



My Hypotheses



Top 5 hypotheses

Loaded from the last completed run (`state.json`); this persists across tab switches and app restarts.

Elo	Matches	Gen	Ranked	Statement
1260	3	1	yes	SEL1L loss-of-function in bortezomib-resistant RPMI-8226 multiple myeloma cells confers resistance specifically by upregulating autophagic flux to clear misfolded proteins
1223	9	0	yes	SEL1L loss-of-function, by redirecting misfolded protein flux toward autophagy rather than proteasomal degradation, confers bortezomib resistance in multiple myeloma
1222	2	1	yes	UBE2S, an E2 ubiquitin-conjugating enzyme that extends K11-linked ubiquitin chains on ERAD substrates, is transcriptionally co-regulated with HSPA5 (GRP78)/BiP
1221	2	1	yes	In bortezomib-resistant myeloma cells, DERL1 upregulation sustains a p97-dependent retrotranslocation route that channels misfolded ER proteins toward lysosomal degradation
1221	1	1	yes	UBE2S sustains bortezomib resistance in multiple myeloma by extending K11-linked ubiquitin chains on misfolded immunoglobulin heavy chain (IgHC), rerouting it to autophagy

▼ #1: Elo 1260 rationale + experiment

Rationale

The original hypothesis correctly identifies autophagy rerouting as a potential escape mechanism upon SEL1L loss, supported by Geneformer-predicted perturbation of RUVBL2, WDR45, and WIPI2 and by published evidence that partial ERAD disruption activates non-proteasomal degradation (PMID 24069311). However, the original statement is underdetermined in three ways that limit falsifiability. First, it does not specify a quantitative threshold for the autophagy flux increase required to claim mechanistic relevance, making it possible to confirm the hypothesis with any detectable change. Second, it does not distinguish whether chloroquine synergy with bortezomib is SEL1L-KO-specific or a general feature of myeloma cells, so a positive result could reflect non-specific lysosomal toxicity rather than rescue of the SEL1L-dependent escape route. Third, it does not disentangle whether the resistance phenotype requires the ERAD-scaffolding function of SEL1L (its role in recruiting misfolded substrates to UBE2S for proteasomal delivery) versus a hypothetical ERAD-independent role (e.g., direct autophagy suppression). The ERAD-dependent paradigm

Cross-check with other models

The Elo leaderboard is the LLM judge's opinion. Before trusting it to pick a wet-lab candidate, cross-check it against **independent quantitative models**: each scoring a hypothesis from a completely different angle. Low correlation = a flag. This is the *panel of judges*.

From your Start here plan, check these:

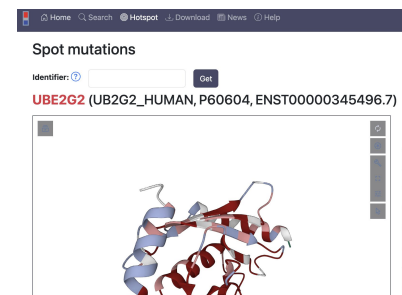
- **HSPA5** → *DepMap* (Assess dependency of myeloma cells on HSPA5 for survival.)
- **SEL1L** → *Open Targets* (Link SEL1L variants to bortezomib resistance in myeloma.)
- **DERL1** → *AlphaMissense* (Evaluate coding variants in DERL1 affecting ERAD function.)
- **UBE2G2** → *AlphaMissense* (Assess missense variants in UBE2G2 altering ubiquitination activity.)

Score a hypothesis

Pick a hypothesis, then run whichever models apply. Each box says what it needs and what it found.

From your leaderboard

SEL1L loss-of-function in bortezomib-resistant RPMI-8226 multiple myeloma cells confers resistance specificall...



HYPOTHESIZE

HYPOTHESES

HYPOTHESIS	RANK
SEL1L loss drives bortezomib resistance	1
Autophagy block re-sensitizes SEL1L cells	2
UBE2S with p97 restores lethality	3

MODEL SCORES

DepMap

OpenTargets

AlphaMissense

Scores from other models, read in alongside the run.

Protocol, reagents, bench.

EXPERIMENT

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Wk. Determine whether autophagy inhibition (chloroquine) vs. residual bortezomib vs. total BPR3-4229-kill is bortezomib-sensitized potential (vs. using BPR3-Blockade (XtremeRNA) as a comparator with to probe BPR3-autophagy control).

Experiments

BPR3-4229/B12 (bortezomib + vehicle) vs. bortezomib/B12, same position same place

Treatment

Bortezomib 0, 1, 2.5, 5, 10, 20 nM • Chloroquine 0, 10, 20 μ M • XtremeRNA 0, 10 μ M
for 48h (or 72h in 2nd day of treatment)

Key comparison: IC50 of bortezomib in BPR3-4229/B12 + chloroquine vs. BPR3-4229/B12 vehicle, same comparison repeated in potential BPR3-4229 to test if the specificity of vs. sensitization

Controls

- Vehicle (BPR3-4229). Baseline viability reference for both cell lines, controls for solvent toxicity
- Chloroquine 25 μ M alone, no bortezomib. Confirms chloroquine single-agent cytotoxicity is separable from combination effect.
- Bortezomib 20 μ M alone, no bortezomib. Confirms BPR3-Blockade single-agent effect, already flagged as weak but needed for factorial randomization.
- Bortezomib 25 nM alone. Positive kill control confirming parental sensitivity and system line resistance.

Readout: XtremeRNA reduction of 70%, converted to % viability in vehicle fit parameters log (logistic curve) pre-treatment is minimal (20% BPR3-4229) + chloroquine reduces bortezomib (bortezomib-kill in BPR3-4229/B12 but not 5-fold in bortezomib/B12); no differential shift, or potential equally to sensitization.

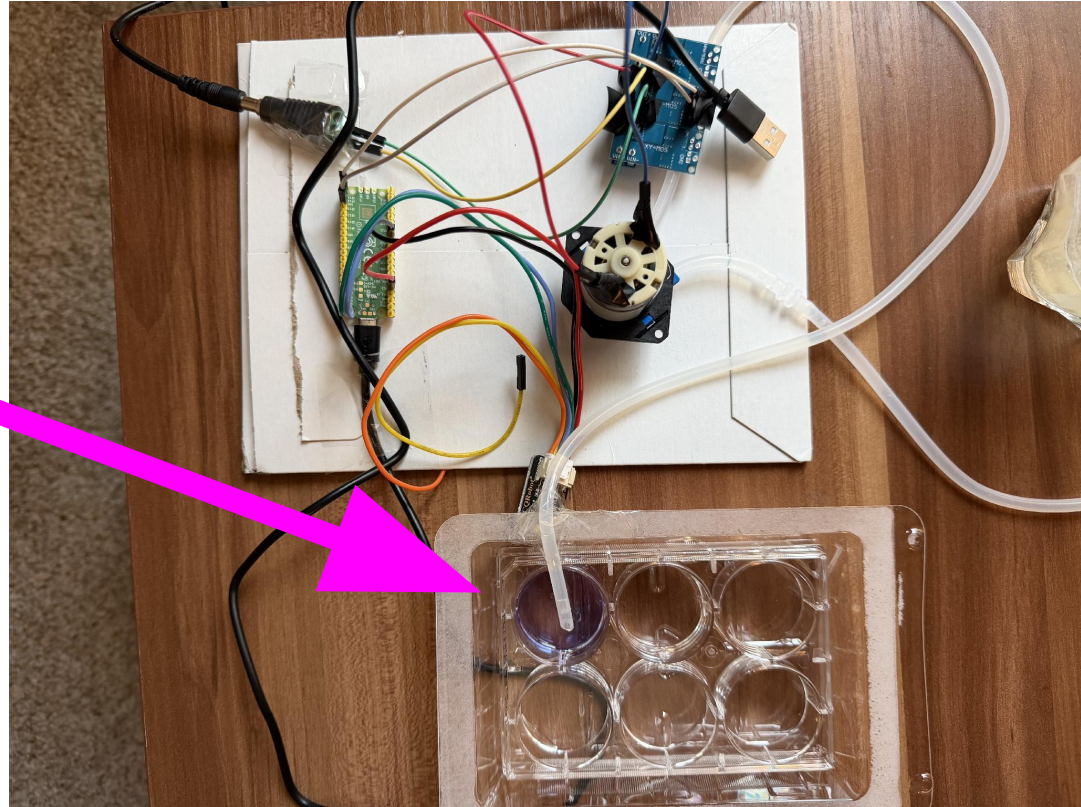
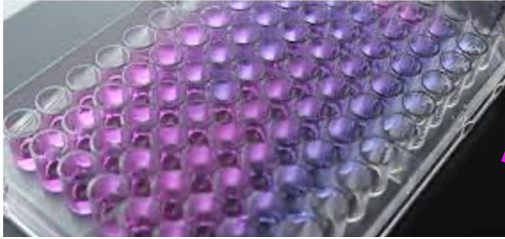
Watch out for Chloroquine is lysosomotropic and raises lysosomal pH (acidic), cytotoxicity in 25 μ M may be autophagy-independent, making any vs. sensitization uninterpretable in autophagy-specific.

What this can't tell us: the genetic manipulation is a positive (vs. BPR3-4229/B12, and BPR3-4229/B12) but will not be a negative (vs. BPR3-4229/B12). Chloroquine is a weak autophagy (only 10% reduction) confirm (vs. BPR3-4229/B12), cannot distinguish BPR3-dependent from non-BPR3-dependent (vs. BPR3-4229/B12), and cannot prove inhibited autophagy is necessary in the substrate. Open Targets (2020) and (2021) scores, plus (2021) very low functional support in myeloma viability, already not reflecting the hypothesis string. The proposed mechanism (bortezomib) only genetic loss of BPR3 (bortezomib) is a target (vs. BPR3-4229/B12) but not a target (vs. BPR3-4229/B12).

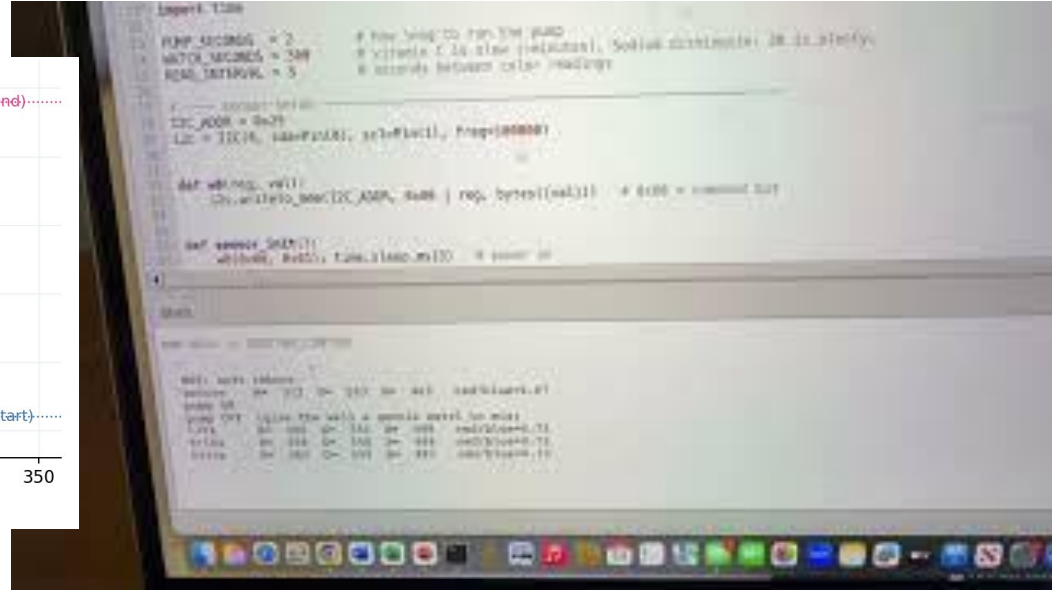
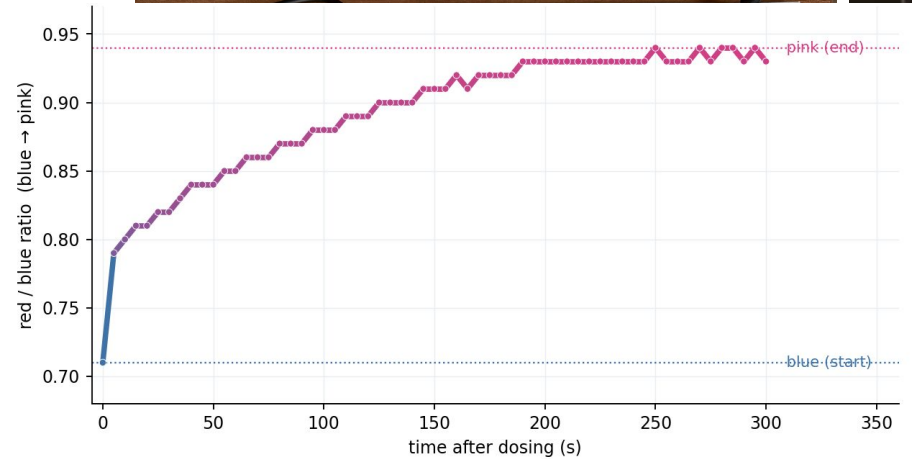
Run the viability experiment



resazurin (blue) → resorufin (pink):
live cells change the color

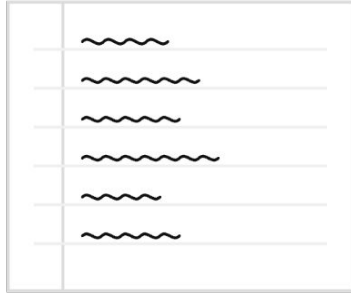


Run the viability experiment

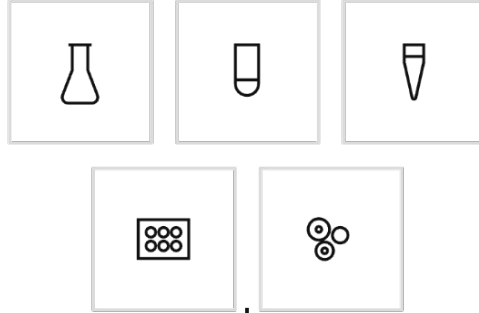


EXPERIMENT

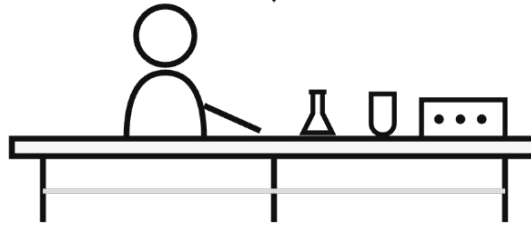
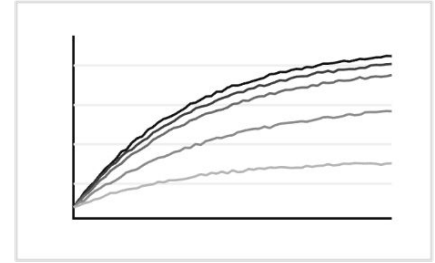
PROTOCOL



REAGENTS



RESULTS

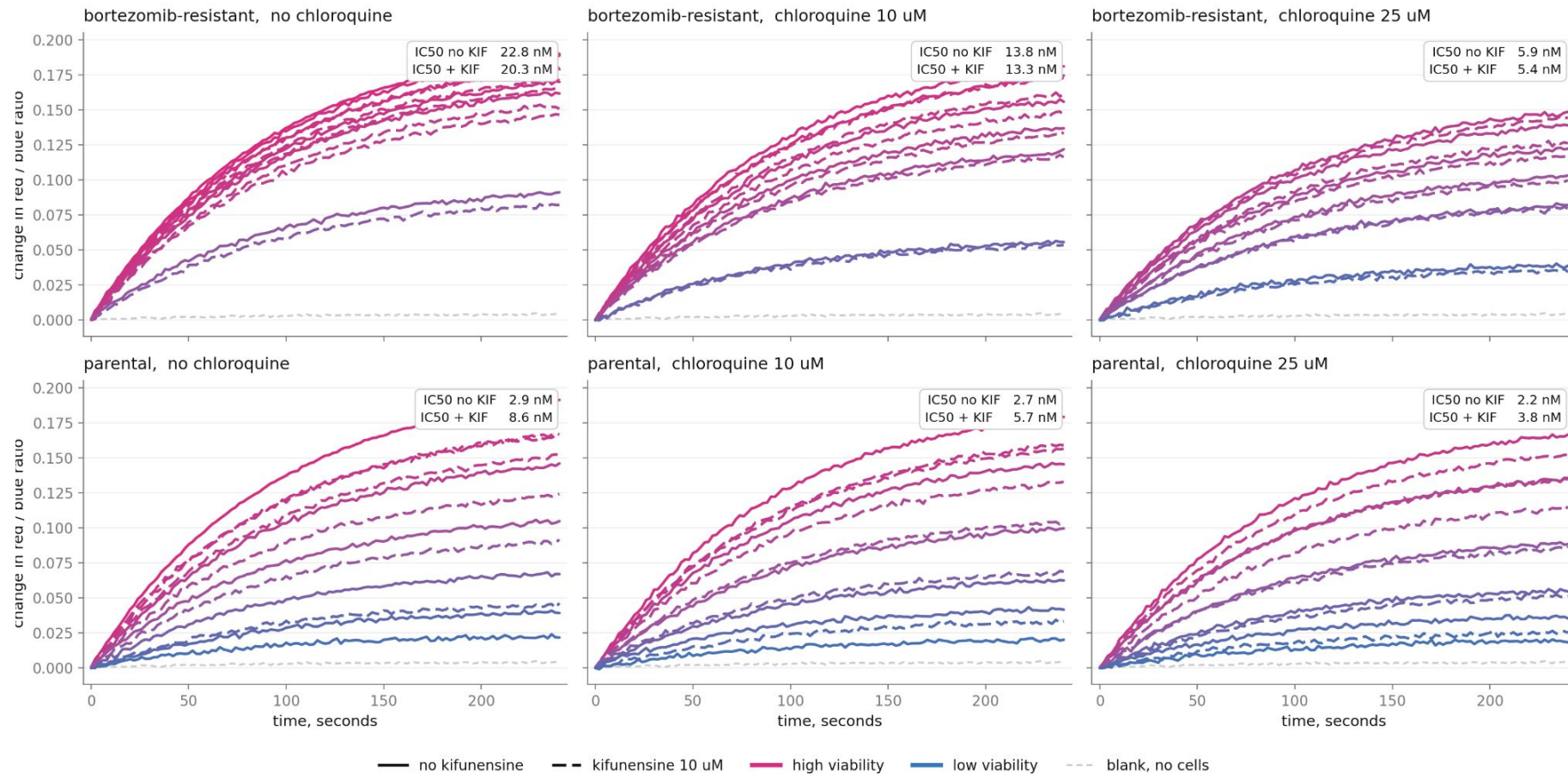


experiment to test my hypothesis

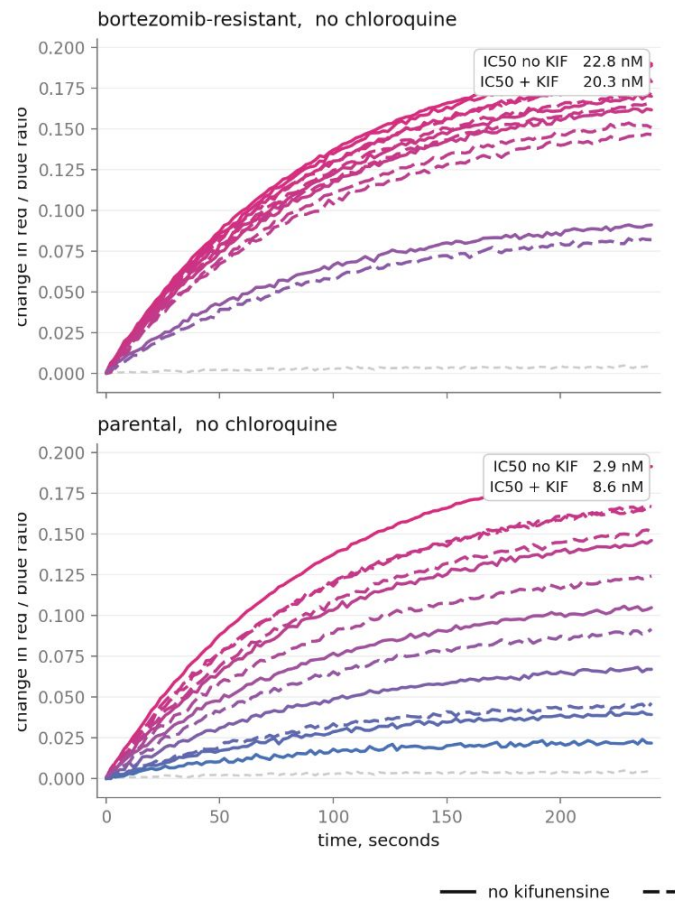
What do the results indicate?

ITERATE

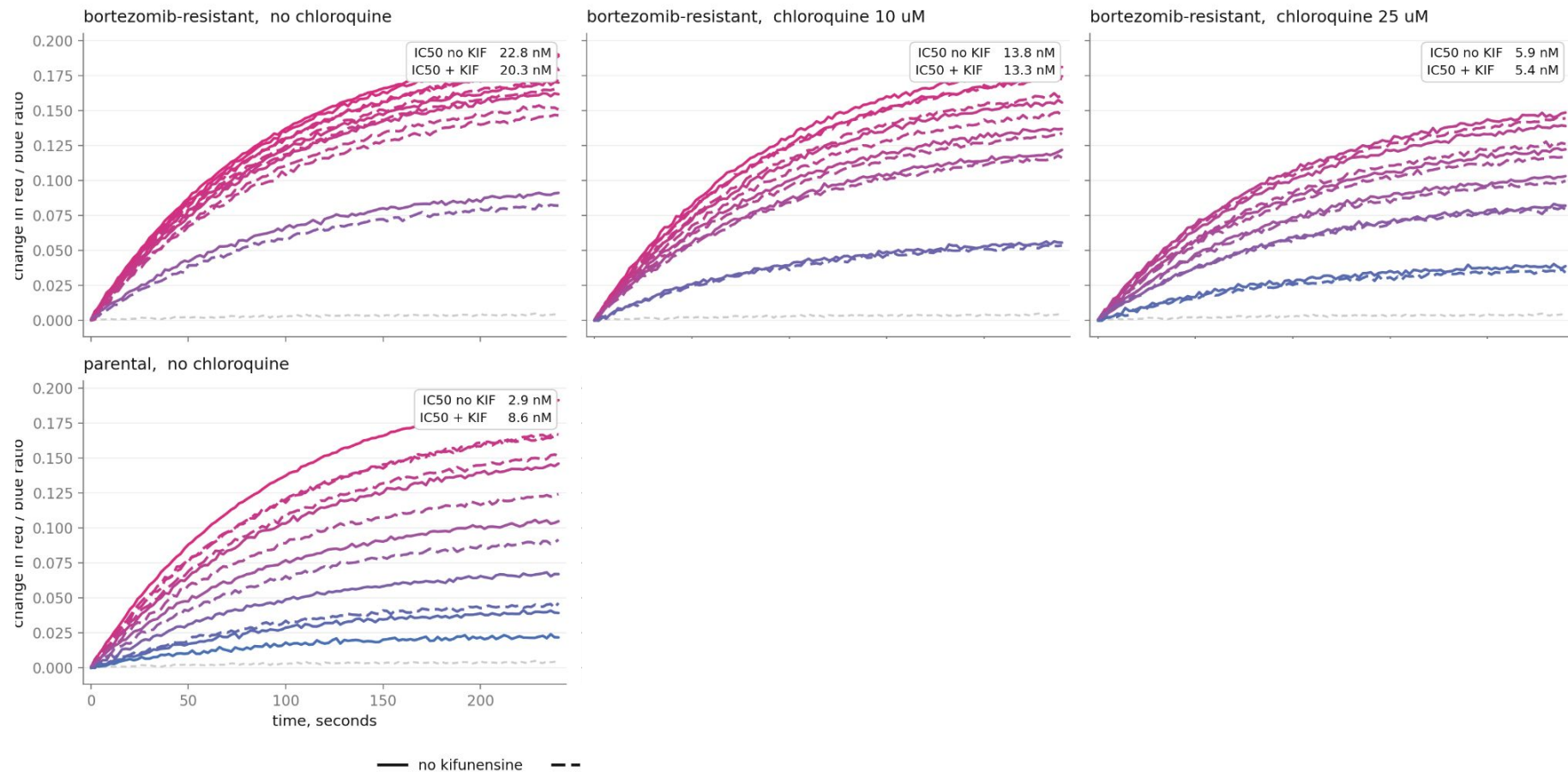
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ix bortezomib doses per panel, 0 to 25 nM, coloured by viability. Simulated plate, 5 x 96-well, blanks subtracted, each line normalised to its own vehicle. Kifunensine shifts parental to the right and leaves the resistant line where it was.



ix bortezomib doses per panel, 0 to 25 nM, coloured by viability. Simulated plate, 5 x 96-well



2. What it means for the hypothesis

Benchmate re-reads the hypothesis against this result and the design that produced it, then says whether it survived.

SEL1L loss-of-function in bortezomib-resistant RPMI-8226 multiple myeloma cells confers resistance specifically by upregulating autophagic flux to clear misfolded immunoglobulins, such that (i) SEL1L-KO cells will show ≥ 2 -fold increase in autophagic flux (LC3-

Feed this back into the hypothesis

supported in part. CQ 25 μM shifted bortezomib IC50 in RPMI-8226/BTZ from 22.8 nM to 5.9 nM (3.9-fold), meeting the ≥ 3 -fold criterion, while the parental shift was 1.3-fold (2.9 $\rightarrow$ 2.2 nM), well below the 1.5-fold threshold for a non-selective effect, satisfying the pre-specified selectivity criterion for the viability prediction. The mechanistic attribution to SEL1L-dependent autophagic flux, LC3-II turnover, ERAD-dead mutant effects, and genetic autophagy inhibition (ATG5-KO) remains entirely beyond this assay's reach.

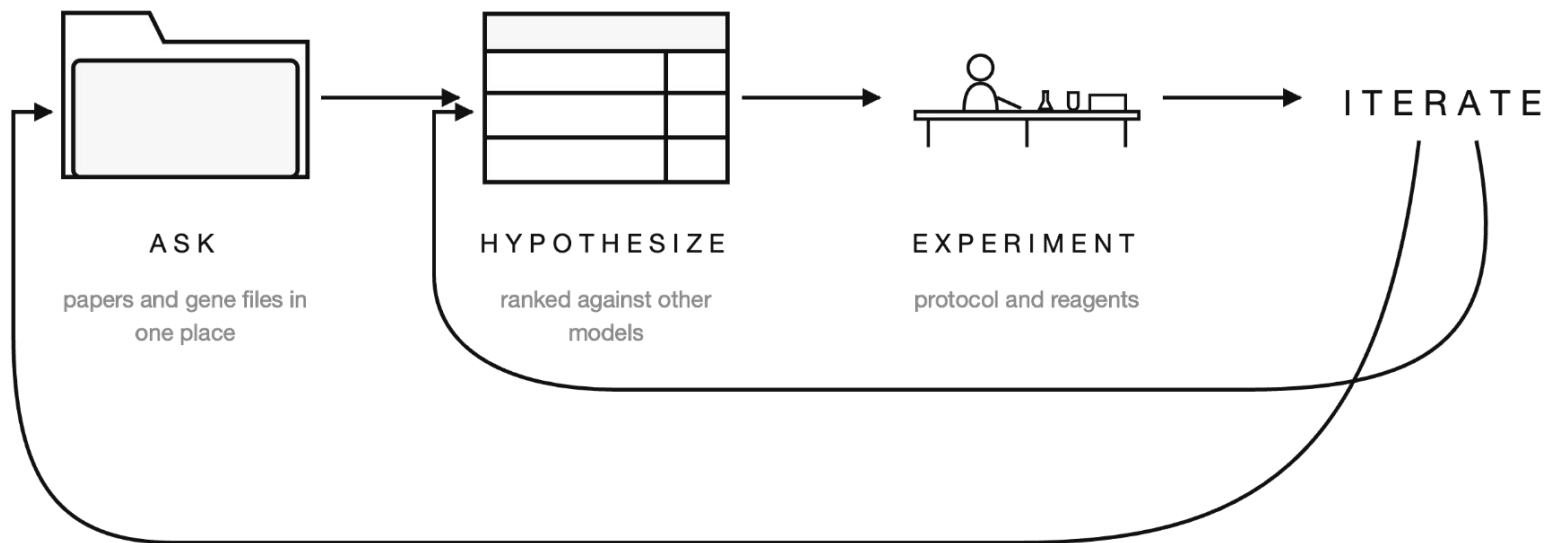
Sharper hypothesis

Chloroquine at 25 μM re-sensitizes bortezomib-resistant RPMI-8226/BTZ cells to bortezomib with a 3.9-fold IC50 reduction (22.8 nM $\rightarrow$ 5.9 nM), meeting the pre-specified ≥ 3 -fold criterion, while producing only a 1.3-fold shift in parental cells (2.9 nM $\rightarrow$ 2.2 nM), confirming selective re-sensitization of resistant cells. The mechanistic claim—that this selectivity is driven by SEL1L loss-of-function upregulating autophagic flux to clear misfolded immunoglobulins—remains entirely untested by this assay. The revised testable prediction is: the chloroquine-selective re-sensitization in resistant cells is autophagy-pathway-dependent rather than a generic lysosomotropic effect, which can be partially interrogated by substituting a more selective autophagy inhibitor (bafilomycin A1) and confirming the IC50 shift pattern replicates, while noting that CQ 10 μM

Next experiment

Run the same full-factorial alamarBlue plate replacing chloroquine with bafilomycin A1 (0, 5, 20, 50 nM) as a more V-ATPase-selective lysosomal inhibitor, crossed with the same bortezomib doses (0, 1, 2.5, 5, 10, 25 nM) in both RPMI-8226 parental and RPMI-8226/BTZ in strict parallel, 72 h, n=3 biological replicates, with DMSO vehicle and bafilomycin-alone controls. If bafilomycin A1 reproduces the ≥ 3 -fold IC50 reduction selectively in resistant cells (without the broad lysosomotropic confound of chloroquine), it strengthens the autophagy-pathway interpretation of the selectivity signal; if it fails to shift IC50 in resistant cells, the chloroquine effect is more likely attributable to non-autophagy lysosomal toxicity, which would weaken the autophagy arm of the hypothesis.

The Loop



**Your AI labmate for every step of the
scientific process.**

ASK

HYPOTHESIZE

EXPERIMENT

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benchmate.streamlit.app

Appendix

Hypothesis

Generation proposes.



the evidence

GENERATION

proposes

REFLECTION

objects

RANKING

scores

PROXIMITY

de-duplicates

EVOLUTION

sharpens

META-REVIEW

catches repeats

SUPERVISOR

decides who runs next

UBE2S extends K11-linked ubiquitin chains on ERAD substrates and acts as a parallel resistance node 1200

Reflection looks for objections.



the evidence

GENERATION
proposes

REFLECTION
objects

RANKING
scores

PROXIMITY
de-duplicates

EVOLUTION
sharpens

META-REVIEW
catches repeats

SUPERVISOR
decides who runs next

UBE2S extends K11-linked ubiquitin chains on ERAD substrates and acts as a parallel resistance node 1200

Other agents refine.



the evidence

GENERATION

proposes

REFLECTION

objects

RANKING

scores

PROXIMITY

de-duplicates

EVOLUTION

sharpens

META-REVIEW

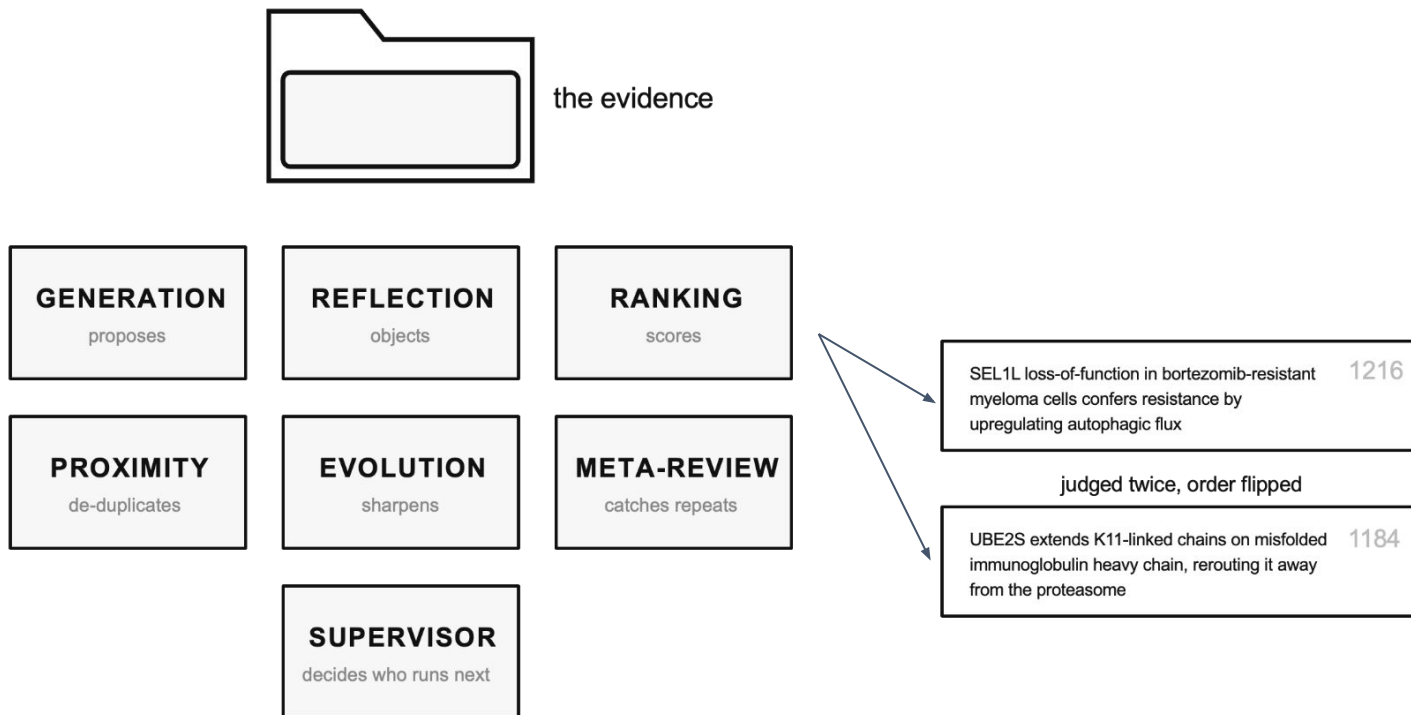
catches repeats

SUPERVISOR

decides who runs next

UBE2S extends K11-linked ubiquitin chains on ERAD substrates and acts as a parallel resistance node 1200

Ranking compares hypotheses.



One match

WHAT THE JUDGE SEES

Statement	the claim itself
Rationale	why the proposing agent thinks it holds
Proposed experiment	first 600 characters of the design
Prior reviewer note	Reflection's most recent objection, if any

THE ORDER SWAP

PASS 1	PASS 2	RECORDED AS
A wins	A wins	A wins, confident
A wins	draw	draw
A wins	B wins	draw, position bias

Language models favor whatever they read first, so every pair is judged twice. The third row does two jobs. It cancels reading-order preference, and it flags pairs too similar to separate, because those are the ones most likely to flip.

What the judge is comparing

THE RUBRIC COMES FROM YOUR QUESTION

The planner reads the research question at the start of the run and pulls evaluation criteria out of it. Those become the rubric. Fallback when nothing is extractable: novelty, plausibility, testability.

TWO HYPOTHESES FROM THE SAME EVIDENCE

VERSION A

SEL1L loss confers bortezomib resistance in RPMI-8226 cells by upregulating autophagic flux.

VERSION B

SEL1L loss confers bortezomib resistance in RPMI-8226 by rerouting misfolded immunoglobulin light chain from the proteasome to autophagy. Predicts **≥2-fold** LC3-II flux increase and **≥3-fold** rightward IC50 shift in SEL1L-KO, with no shift in parental.

The judge picked B. It names the substrate being rerouted and states thresholds the experiment can miss.

WHY A CANNOT FAIL

Autophagic flux is never exactly zero, so any detectable increase reads as support. Run the experiment and you confirm the hypothesis whatever comes back.

WHERE B CAN FAIL

IF THE RUN SHOWS	VERSION B IS
1.4-fold flux increase	refuted, below the 2-fold call
2-fold IC50 shift	refuted, below the 3-fold call
parental line also shifts	refuted, not SEL1L-specific

The plate came back at **2.7-fold**. Below Version B's own threshold. The hypothesis failed its own test, which was only possible because the number was written into it.

How ratings move

1200

every hypothesis starts here

THE ARITHMETIC

```
expected = 1 / (1 + 10 ^ ((R_opp - R_self) / 400))
new      = old + K * (actual - expected)
actual   = 1 win    0.5 draw    0 loss
```

expected is the win probability implied by the two current ratings alone. A 200-point gap gives the favourite about 76 percent. Equal ratings give 50 percent. The rating then moves by however much the result surprised us.

WHAT K IS

K is the ceiling on how much one match can move a rating.

(actual – expected) can never exceed 1, and only approaches 1 when the outcome was maximally surprising. So the largest possible swing from a single match is K points.

MATCHES PLAYED	K	WHY
under 5	40	the rating is still a guess, let evidence move it
5 to 15	20	it is settling
over 15	10	many judgments are baked in, one more should not overturn them

FOUR WORKED MATCHES

Two brand-new hypotheses, both 1200. A wins.

```
expected for A = 1 / (1 + 10 ^ (0/400)) = 0.500
K for both      = 40
A: 1200 + 40 * (1 - 0.500) = 1220
B: 1200 + 40 * (0 - 0.500) = 1180
```

A 40-point gap from one match, because nothing was known about either.

The underdog wins. A 1150 with 2 matches, B 1350 with 6.

```
expected for A = 1 / (1 + 10 ^ (200/400)) = 0.240
K for A = 40      K for B = 20
A: 1150 + 40 * (1 - 0.240) = 1180.4 (+30.4)
B: 1350 + 20 * (0 - 0.760) = 1334.8 (-15.2)
```

A big move for the winner nobody expected. A smaller one for the better-established rating.

The favorite wins. A 1350 with 20 matches, B 1150 with 16.

```
expected for A = 1 / (1 + 10 ^ (-200/400)) = 0.760
K for both      = 10
A: 1350 + 10 * (1 - 0.760) = 1352.4 (+2.4)
B: 1150 + 10 * (0 - 0.240) = 1147.6 (-2.4)
```

Barely anything happens. The result told us nothing new.

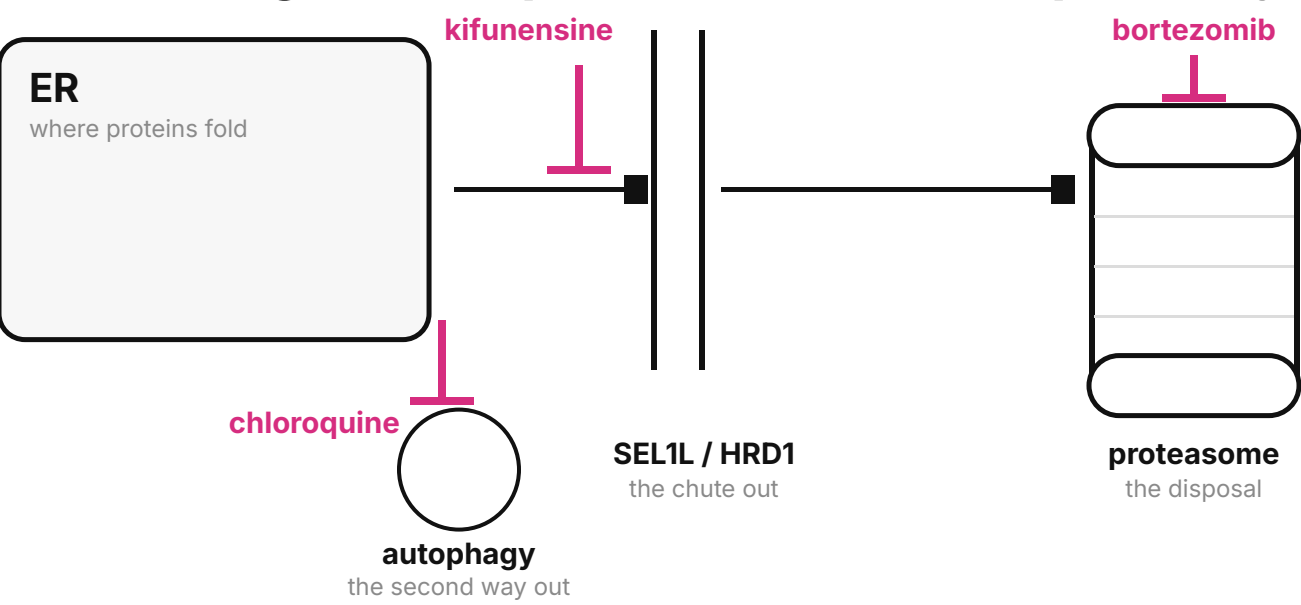
The draw that costs you. A 1300, B 1100, both 8 matches.

```
expected for A = 0.760
K for both      = 20
A: 1300 + 20 * (0.5 - 0.760) = 1294.8 (-5.2)
B: 1100 + 20 * (0.5 - 0.240) = 1105.2 (+5.2)
```

A drew with a hypothesis rated 200 below, so A loses points. This is what punishes hypotheses that cannot distinguish themselves. Near-identical claims flip on the order swap, score a draw, and the higher-rated one bleeds.

Experiment

Three drugs, three points on the same pathway



DRUG	BLOCKS	IN PLAIN WORDS	WHY IT IS HERE
bortezomib	the proteasome	jams the shredder	the frontline myeloma drug, and the one resistance is against
kifunensine	ER mannosidase I	stops garbage reaching the chute	the closest drug to SEL1L, the gene in the hypothesis
chloroquine	lysosomal acidification	blocks the second way out	tests whether autophagy is the escape route

Why kifunensine stands in for SEL1L

APPENDIX

ERAD IS TWO STEPS

DECIDE — is this protein terminally misfolded?

mannose trimming acts as a clock. EDEM1 and EDEM3 read the trimmed glycan and hand the protein to SEL1L-HRD1.



kifunensine

DESTROY — pull it out and break it down

SEL1L-HRD1 extracts it to the proteasome.



bortezomib

THE MEASURED LINK

“Kifunensine inhibited the binding of EDEM1 and EDEM3 with SEL1L.”

Saeed et al., J Biol Chem 2011, 10.1074/jbc.M111.259085

So kifunensine blocks the handoff into SEL1L.

THE PREDICTION IT MAKES

If losing SEL1L function causes bortezomib resistance, then blocking that step with a drug should make sensitive cells resistant, and should do nothing to cells that have already lost it.

WHAT THE PLATE RETURNED

parental, bortezomib IC50 with kifunensine

2.9 → 8.6 nM, 3.0x more resistant

resistant, bortezomib IC50 with kifunensine

22.8 → 20.3 nM, no change

Both directions came out as predicted. Kifunensine plus bortezomib returns zero papers on PubMed, so this arm is unstudied.

Iterate

How to read the grid

APPENDIX

THE LAYOUT

left to right along a row	more chloroquine: 0, then 10, then 25 μ M
top row to bottom row	swap cell line: resistant, then parental
solid vs dashed in a panel	kifunensine off, then on at 10 μ M
six curves per panel	six bortezomib doses: 0, 1, 2.5, 5, 10, 25 nM
flat grey dashes	blank wells, no cells, the zero line

12 curves per panel. 72 in total.

EVERY IC50 ON THE FIGURE, IN nM

	no CQ	CQ 10 μ M	CQ 25 μ M
resistant, no KIF	22.8	13.8	5.9
resistant, + KIF	20.3	13.3	5.4
parental, no KIF	2.9	2.7	2.2
parental, + KIF	8.6	5.7	3.8

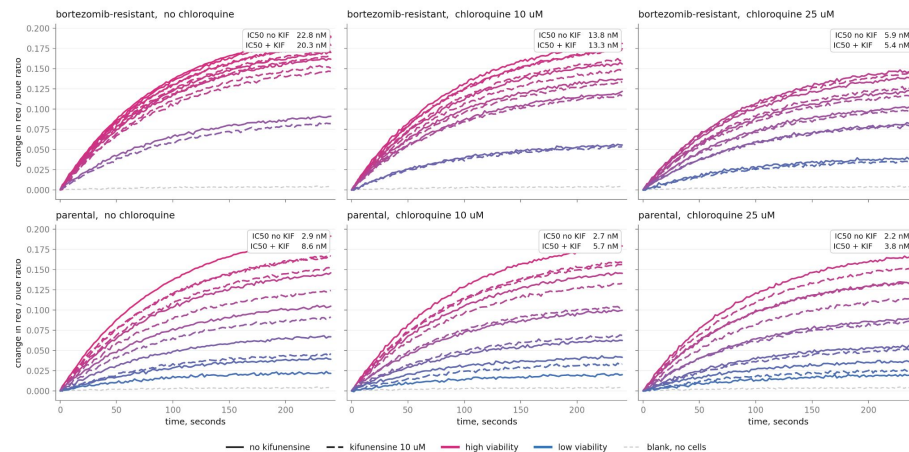
Those four numbers carry the whole result.

THE DIRECTION TRAP

- curve **HIGHER** on the plot $\rightarrow$ more cells alive
- IC50 **LOWER** $\rightarrow$ less drug needed to kill half, **MORE sensitive**
- IC50 **HIGHER** $\rightarrow$ more drug needed, **MORE resistant**

A smaller IC50 feels like a smaller effect and it is the reverse. 5.9 nM means those cells die at a quarter of the dose they used to need.

Complete factorial: bortezomib x chloroquine x kifunensine, both cell lines, n=3



A7

nk bortezomib doses per panel, 0 to 25 nM, coloured by viability. Simulated plate, 5 x 96-well, blanks subtracted, each line normalised to its own vehicle. Kifunensine shifts parental to the right and leaves the resistant line where it was.

Top row: the resistant line

APPENDIX

NO CHLOROQUINE

IC₅₀ **22.8** nM without kifunensine, **20.3** with.

Every curve bunched high. Across a 25-fold bortezomib range these cells barely respond, and only the 25 nM arm drops. Solid and dashed lie on top of each other.

TWO JOBS: this is the baseline every other resistant panel is measured against, and it is the negative control for kifunensine. If kifunensine works by blocking the SEL1L step, and this line has already lost SEL1L function, it should be inert here. It is.

CHLOROQUINE 10 UM

IC₅₀ **13.8** nM. A **1.7-fold** shift.

Everything comes down and the curves spread apart. Real, and under the 3-fold bar the hypothesis set. This panel is what says the effect is dose-dependent rather than all-or-nothing.

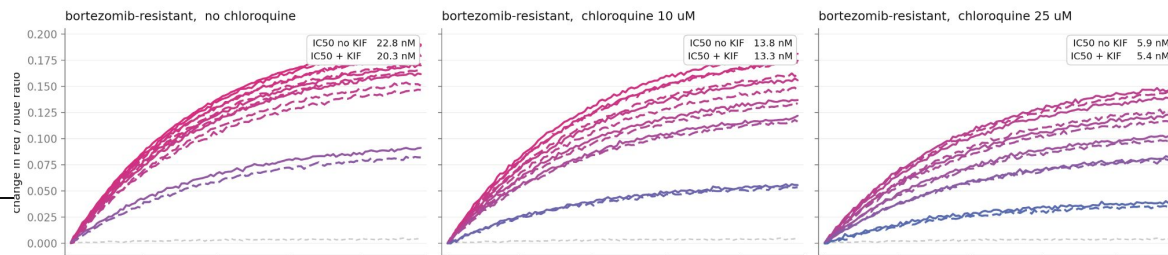
CHLOROQUINE 25 UM

IC₅₀ **5.9** nM. A **3.9-fold** shift.

The headline. Curves much lower, the bottom ones approaching the blank. Clears the pre-specified 3-fold criterion.

Kifunensine adds nothing in all three panels: 20.3, 13.3, 5.4 against 22.8, 13.8, 5.9. Consistent across the row rather than one lucky panel.

Complete factorial: bortezomib x chloroquine x kifunensine, both cell lines, n=3



Bottom row: the parental line

APPENDIX

NO CHLOROQUINE

IC50 **2.9** nM without kifunensine, **8.6** with. **3.0-fold MORE resistant**.

The dashed curves sit ABOVE the solid ones at matched bortezomib dose. Higher means more alive. Kifunensine is protecting these cells: they now need three times more bortezomib for the same kill.

THE NOVEL RESULT. Blocking the SEL1L step with a drug makes sensitive cells behave like resistant ones. Kifunensine plus bortezomib returns zero papers on PubMed.

Also here: parental 2.9 nM against resistant 22.8 nM gives a 7.9-fold resistance factor for the line.

CHLOROQUINE 10 UM

IC50 **2.7** nM without kifunensine, **5.7** with.

The epistasis panel, and the subtle one. Read it against the panel to its left.

without kifunensine,	2.9 → 2.7
chloroquine does nothing:	nM
with kifunensine on board, it works:	8.6 → 5.7 nM

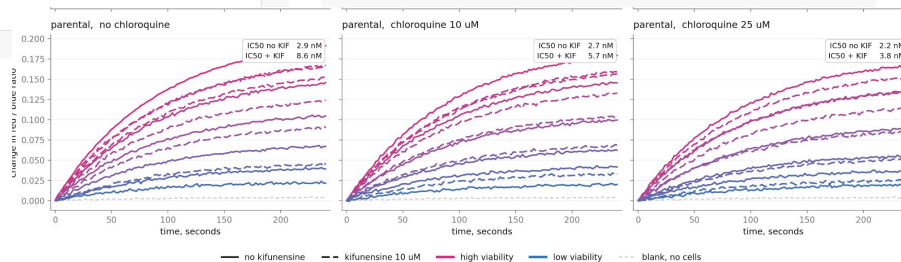
Chloroquine only matters once the ERAD step is blocked. The autophagy route is not load-bearing in a cell whose ERAD works. Same statement whether the block is genetic, as in the resistant line, or pharmacological, as here.

CHLOROQUINE 25 UM

IC50 **2.2** nM. A **1.3-fold** shift.

The selectivity control

Against the top-right panel: the same chloroquine dose moves the resistant line 3.9-fold and this one 1.3-fold, below the 1.5-fold threshold. The re-sensitisation is specific to the resistant cells.



1x bortezomib doses per panel, 0 to 25 nM, coloured by viability. Simulated plate, 5 x 96-well, blanks subtracted, each line normalised to its own vehicle. Kifunensine shifts parental to the right and leaves the resistant line where it was.